CIS 5450: Big Data Analytics
Fall 2026, Online
Prof. Zachary Ives, Prof. Nan Zheng

# Big Data Analytics: Final Project

## Description of the Project

The objective of this project is for you to demonstrate that you are able to apply the topics taught in this course to a real-world setting. You are welcome to use topics and concepts beyond the scope of the course so long as they are adjacent to the concepts taught in this course. This project is a great opportunity to create a "portfolio artifact" that you could add to your resume, post online, share on LinkedIn, or discuss during a job interview.

Projects will be done in groups of 3 people - **no exceptions**. Although not mandatory, it would be best to have all of the team members in time zones where you all can work together without problems. **Each team member should have clearly defined and agreed-upon roles and responsibilities.**

Each group will be assigned to a **Project TA.** Your Project TA is your mentor for the project and is also responsible for grading your project. You are **strongly encouraged** to attend your TA's office hours. You should reach out to them directly (via private Ed Discussion posts or email) if you have questions or would like to schedule a time to meet. Your TA mentor will be assigned **approximately 1 week before proposals are due.**

In addition, you will be **required** to schedule and attend an intermediate check-in with your TA mentor by the deadline listed below.

You can either (1) choose a staff-provided dataset or (2) to propose your own dataset/project. Either way, each group must submit a project proposal (more information below).

Your project is expected to have the following components, at minimum:

1. Introduction / Background
2. EDA (Exploratory Data Analysis)
3. Machine Learning / Statistical Modeling
4. Description of Challenges / Obstacles Faced / Lessons Learned
5. Potential Next Steps / Future Direction

Please make sure to submit the names of your teammates and indicate any preferences you have for your TA mentor in Gradescope by **10/19 at 11:59 PM ET.**

Mentors will be assigned on **10/20.**

This project has 4 deliverables in total:

1. Proposal: **DUE Monday October 26th at 11:59 PM ET**
2. Intermediate Check-in: **DUE Sunday November 8th at 11:59 PM ET**
3. Notebook containing all code and annotations: **DUE Wednesday December 2nd at 11:59 PM ET**
4. Project Video Presentation: **DUE Wednesday December 2nd at 11:59 PM ET**

Logistical Notes:

- This is a 3-person group project. No collaboration outside your group is allowed.
- **Your notebook should *not* be based on prior work completed outside the class, by yourself or by others.**
- Please use the **designated / pinned Ed Discussion post** to find group members, taking into account potential time zone differences.
- **There are no late days or extensions for any component of the project.**

## Group Sign-up and TA Mentor Request Assignment: DUE Monday October 19th at 11:59 PM ET

Each group is **required** to submit the following to Gradescope:

1. A list of all your group members' names
    a. **We encourage you to use the pinned thread on Ed Discussion to form teams.**
2. A list of all your group members' email addresses

3. TA name if you would like to request a TA — note that groups who submit earlier rather than later and those who submit a proposed project title and dataset will have a greater chance of getting assigned to their preferred mentor. These will not be guaranteed though.
    a. If you do not have a preference, you can select all of the TA names in Gradescope.
    b. If you have several preferences, you may also indicate this; however, you will only be assigned one mentor.

Gradescope is set up to accept group submissions, so there is **no need for all** team members to submit this assignment. That being said, for students submitting on behalf of their group, please make sure to add **all** of your teammates to your Gradescope submission.

Please make sure to fill out this Google Form as well (also available in Canvas):
https://forms.gle/JH3Je9JSeSRtFirq9

## Proposal Guidelines: DUE Monday October 26th at 11:59 PM ET

Each group is required to submit a **project proposal** by the date indicated above. This is required for all groups, including those using staff proposed datasets.
Your proposal should be roughly **300 words** in total and contain the following:

1. All group member's names
    a. Breakdown of what each member will be responsible for
2. Data source - if you are proposing your own dataset / topic.
    a. Good data sets (i.e., those where you can actually build useful and interesting models) should have:
        i. >= 50,000 rows **after cleaning and dropping null values**
        ii. A rich set of features / properties that intuitively should be useful in predicting outcomes.
3. A count of the total number of rows in your dataset **after cleaning and dropping null values**
4. Project Plan
    a. Explain what you intend to study with your project.
    b. What is the ultimate objective?
    c. What types of models are you considering?

5. Why is this project interesting?
6. What challenges and obstacles might you anticipate with this project?

While your project can certainly vary from the proposal that you submit, the proposal component is designed to help make sure that you are on the right track and so you can receive feedback as you go.

Gradescope is set up to accept group submissions, so there is **no need for all** team members to submit the proposal. That being said, for students submitting on behalf of their group, please make sure to add **all** of your teammates to your Gradescope submission.

## Intermediate Check-in: DUE Sunday November 8th at 11:59 PM ET

During the intermediate check-in week, each team will schedule a Zoom meeting with their project TA to discuss the work they have done on the project and work still left to accomplish. This can either be during the TA's office hours or during an agreed upon time with all group members and your assigned TA.

By this point, we expect you to have completed the EDA and data wrangling part of the project as well as have thought about the modeling process. We will be making sure you are on the right track and answering any questions you may have about your specific project.

**Important note**: In addition to the intermediate check-in, we **strongly** encourage groups to schedule an initial meeting with their TA mentors before the official intermediate check-in. This early conversation will allow TA mentors to provide guidance, identify potential issues, and give your group time to address them before the project moves further along.

## Project Rubric for Deliverable 1

| Criteria | Description | Points ( /105) |
|---|---|---|
| Project Proposal/ Intermediate Check-in | A brief high-level description of what you plan to do and what are your plans for the project | 5 |
| Difficulty | Have you attempted challenging analysis? How much time would have been required to complete your project? | 10 |
| Code Quality/ Readability | Is your project notebook understandable and fairly well broken into modular steps? | 10 |
| Creativity/Uniqueness | Does your project stand out from the rest? Think about how you could make your project relevant and try to perform analysis that is not obvious or has not already been done. | 10 |
| Visualization | Visualize your findings well using plots and graphs. Make sure that they are informative and appealing. We encourage exploring packages such as tensorboard, plotly, seaborn, on top of the traditional matplotlib. | 10 |
| Modeling | Is your model useful? Is your model implemented correctly? The models that you choose should be justified. You are encouraged to explore and implement more than one modeling method. | 20 |
| Application of Course Topics | Is your project built around the topics discussed in class, recitation, or covered on the homeworks? You are welcome to go beyond the scope of the course, but you should apply a significant amount of the course topics. | 20 |
| Quality of Deliverable | Presentation matters! Is your final product clean and polished? Have you created a deliverable that is engaging and informative? Is your notebook well organized, clearly annotated, and easy for a reader to follow from analysis to findings? | 20 |

*Students may fill out a Project Contribution form at the end of the project to discuss each group member's contributions as well as any additional information we need to know. This is not required.

**Please note that your final project dataset must have >= 50,000 rows after cleaning and dropping null values. Projects that do not fulfill this requirement will be subject to a significant penalty.**

In addition, your final project dataset **cannot** be composed of synthetic or fake data. **Projects that use synthetic or fake data will also be subject to a significant penalty.**

The rubric for the video demo can be found in the corresponding document.

## Final Project Deliverables: DUE Wednesday December 2nd at 11:59 PM ET

Each group must submit the following:

Proposal and Check-in:

- See the guidelines and deadlines for these deliverables above [Page 3 and 4].

**Deliverable 1 (worth 20% of the grade):**

- Every group **must** submit a complete notebook or notebooks with all of the code used throughout the project.
- You can submit just 1 notebook, so long as this one notebook contains all of the code that you used for your project.
- Alternatively, you can submit multiple notebooks. I.e., have one that contains all of your code, then make a copy and in the second notebook, remove any uninteresting parts/ code, add analysis, and clean it up.
- This notebook should be reasonably organized and the code should be readable (your TA should be able to understand what you are doing in any part).
- This notebook should also be self-contained in case your TA mentor needs to run it for grading. This means the following:
    - The data you used should be accessible to your TA mentor
    - All necessary packages, installations, etc. should be present in the notebook
    - The ingestion pipeline should be built such that the data is **easily** accessible
- At least one of your notebooks should be annotated regardless if you are submitting one or multiple notebooks

- Think of a notebook similar to the homework assignments.
- Your notebook would be broken up into separate sections and should be **very readable.**
- Each section should have an introduction and some description of the findings. You should describe the analysis that you are performing and the results. Interpret any important visualizations.
- You may submit separate notebook(s) containing the remainder of your code that does not make it into your final annotated notebook.

Gradescope is set up to accept group submissions, so there is **no need for all team members** to submit the proposal.

**Deliverable 2 (worth 5% of the grade)**:
An up to seven minute long video presentation walking through your project. Please see the video instructions linked here (also linked in Canvas). You will submit a link to your video through Canvas. Only one team member needs to submit the link.

**Please be aware that the TAs must be able to open and view your video presentation, so please double check the video permissions. If a TA is unable to view your video for whatever reason, you will not receive credit for that component of the project.**

## Course Staff Proposed Datasets

1. arXiv Dataset - For nearly 30 years, ArXiv has served the public and research communities by providing open access to scholarly articles, from the vast branches of physics to the many subdisciplines of computer science to everything in between, including math, statistics, electrical engineering, quantitative biology, and economics.
2. Health Nutrition and Population - HealthStats provides key health, nutrition and population statistics gathered from a variety of international sources. Themes include population dynamics, nutrition, reproductive health, health financing, medical resources and usage, immunization, infectious diseases, HIV/AIDS, DALY,

population projections and lending. HealthStats also includes health, nutrition and population statistics by wealth quintiles.

3. COVID-19 Open Research Dataset - CORD-19 is a resource of over 200,000 scholarly articles, including over 100,000 with full text, about COVID-19, SARS-CoV-2, and related coronaviruses. This freely available dataset is provided to the global research community to apply recent advances in natural language processing and other AI techniques to generate new insights in support of the ongoing fight against this infectious disease.
4. Open Food Facts - The Open Food Facts database contains nutritional information from foods all over the world. Information includes ingredients, allergens, nutrition facts and all the tidbits of information we can find on product labels.
5. Earth Surface Temperature Data - The Berkeley Earth Surface Temperature Study combines 1.6 billion temperature reports from 16 pre-existing archives. It is nicely packaged and allows for slicing into interesting subsets (for example by country). They publish the source data and the code for the transformations they applied. They also use methods that allow weather observations from shorter time series to be included, meaning fewer observations need to be thrown away.
6. Financial Markets Data - The first four (4) files below contain a combined total of approximately 50 GB of ***nanosecond-resolution*** market by order[1] data for **futures contracts** traded on the Chicago Mercantile Exchange (CME). The contracts include crude oil futures, U.S. Treasury futures, and foreign exchange futures. The data spans from November 15, 2025 to January 15, 2026. Because of the large size of the data, you may need to use a scalable data-processing stack similar to the one you developed in Homework 3. Please plan your workflow accordingly.

   The fifth file below differs from the first four files. It contains ***daily-resolution*** Open, High, Low, Close, Volume (OHLCV) data on futures contracts traded on the CME. The file is approximately 58 MB and spans the timeframe from June 2010 through October 2025. It covers a very broad range of asset classes, including interest rates, crude oil, natural gas, refined products, foreign exchange, agriculture, metals, and digital currencies.

   You may use some or all of the datasets in your analysis.

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[1] Market by Order data is often referred to as "L3 data". See MBO and MBO for additional details.

Please note that financial markets data is inherently noisy and notoriously difficult to forecast. Grading will therefore emphasize the quality of your methodology, data handling, modeling choices, and reasoning rather than raw predictive accuracy.

**Nanosecond-Resolution**:

1. 11/15/2025 to 11/30/2025 (~17.5 GB)
2. 12/01/2025 to 12/15/2025 (~5.5 GB)
3. 12/16/2025 to 12/31/2025 (~6.2 GB)
4. 01/01/2026 to 01/15/2026 (~20.3 GB)

**Daily-Resolution:**

5. 06/10/2010 to 10/31/2025 (~58 MB)

7. NYC Taxi and Limousine Commission (TLC) Trip Record Data - Yellow and green taxi trip records from New York City. It includes fields capturing pickup and drop-off dates/times, pickup and drop-off locations, trip distances, itemized fares, rate types, payment types, and driver-reported passenger counts in addition to other data.

**Please get started early, and feel free to come to office hours or post on Ed Discussion with any questions. Good luck!**